

Did hydration breaks change the 2026 World Cup?

A reproducible analysis of 203 mandatory breaks across 102 matches

Version 2.0.0 · 2026-07-27 · github.com/rezaba002/WC2026-HydrationBreakAnalysis

Every figure in this report is generated by a script from audited source data. The analysis specification was frozen before any outcome was computed; every subsequent amendment, deviation and limitation is dated and justified in `CHANGELOG.md` (A1–A7, E1).

1. The question

The 2026 World Cup was the first to mandate hydration breaks in **every match regardless of weather** — roughly three minutes near the 22nd minute of each half. Previously, cooling breaks were triggered only by a heat threshold. Football was, in effect, divided into four quarters.

The breaks were immediately controversial. Managers said they changed the identity of the game. Commentators narrated momentum swings around them. Broadcasters sold them.

This report asks one question:

Did the universal hydration-break policy measurably change attacking performance — and if the average effect was small, why were the breaks so widely experienced as decisive?

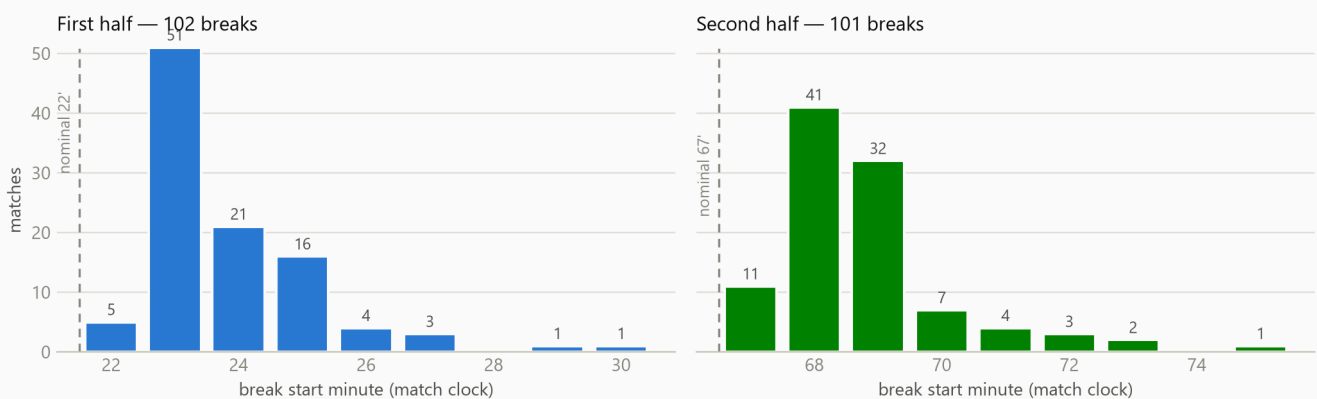
The interesting possibility is not that one side of that is wrong. It is that both are true at once: a near-zero average effect coexisting with real, measurable changes to how football was coached, staffed, timed and narrated.

2. What the breaks actually looked like

Before testing anything, the basic facts.

Hydration breaks started later than the nominal minute

Actual start minutes of the 203 recorded breaks, WC2026 (102 matches). Modal starts: 23' and 68'.

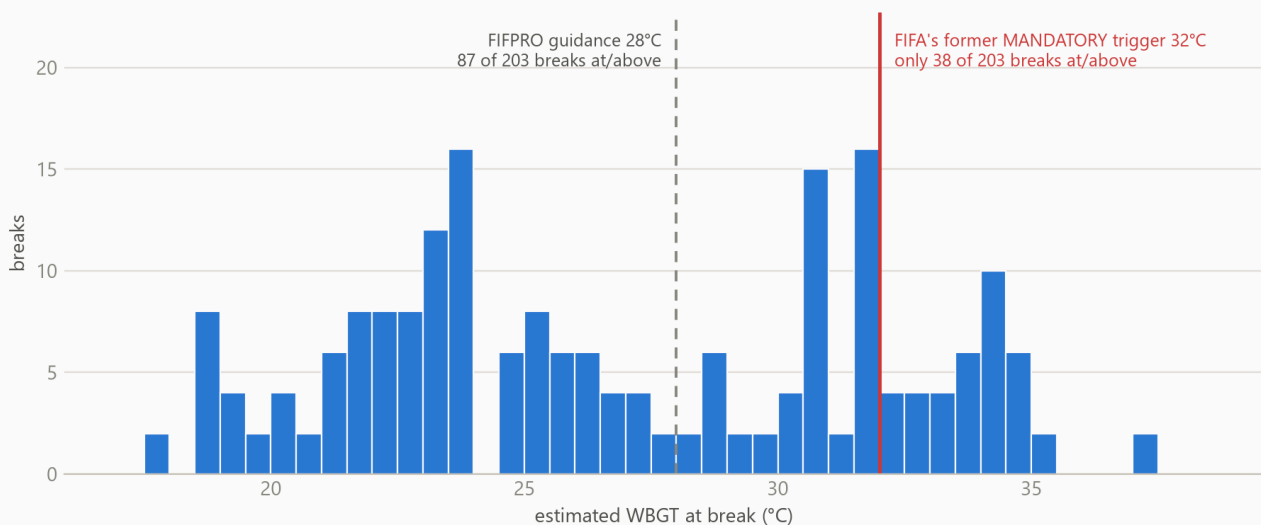


Breaks did not happen at the nominal minute. They clustered at **23'** and **68'**, with a tail out to 30' and 75' — referees waited for a natural stoppage. Of 203 recorded breaks, 102 were first-half and 101 second-half. (One half, France–Iraq's second, has no recorded break in any source we hold; it is documented, not silently dropped.)

This matters for measurement: an analysis anchored to 22'/67' would misplace most breaks. All results here use each match's **actual** break minute.

Four in five breaks happened below FIFA's own former mandatory threshold

Estimated wet-bulb globe temperature at each of the 203 breaks. Median 26.1°C. FIFA and FIFPRO thresholds differ — see report §2.



The heat context explains why the policy was contested. Estimated wet-bulb globe temperature at the breaks ranged from 17.8 °C to 37.2 °C, median 26.1 °C.

Two different thresholds matter here, and they belong to two different bodies [3][4]. Secondary reporting places FIFA's former mandatory cooling-break trigger at **WBGT ≥ 32 °C**, with breaks merely requestable between roughly 27 °C and 32 °C; we did not obtain the protocol document itself, so this figure is cited from press coverage rather than verified at source (see note [3]). FIFPRO, the players' union, recommends breaks from about 26 °C and argues for delaying matches above 28 °C. Against those benchmarks:

threshold	breaks at or above	share
FIFPRO break guidance, 26 °C	103	51%
FIFPRO delay guidance, 28 °C	87	43%
FIFA former mandatory trigger, 32 °C	38	19%

So **four out of five 2026 breaks happened in conditions below that reported former mandatory trigger**. This is a policy change, not a weather event — which is precisely why its effects are worth measuring.

3. How the effect was measured

Two design decisions carry the whole analysis.

The clock problem. The match clock runs during a break. A naive "10 minutes before versus 10 minutes after" comparison therefore gives the *after* side about three fewer minutes of actual football. Every result below is computed on two clocks: the **display clock** (what viewers and event feeds see) and the **break-adjusted clock** (hydration dead time removed — and only that; throw-ins and VAR remain, so it is not an "active play" clock and is never described as one).

The counterfactual problem. Comparing before-and-after tells you nothing on its own, because football is volatile: dominance decays, pressure reverts to the mean. The test must be *versus what normally happens*. For each real break we enumerate eligible non-break minutes in the same match and half, matched on score state and stage, excluding minutes near goals, red cards, VAR reviews and half boundaries — then draw 10,000 randomized pseudo-breaks and compare the real result against that null distribution.

Primary outcome is **balance disruption**: the absolute change in shot differential across the break, $|\Delta(\text{home} - \text{away})|$. It is direction-agnostic, so it can be scored for every break, and it is always defined — unlike shot *share*, which is 0/0 in the many quiet windows. Uncertainty is a match-level bootstrap (5,000 iterations, fixed seed), because two breaks per match are not two independent observations.

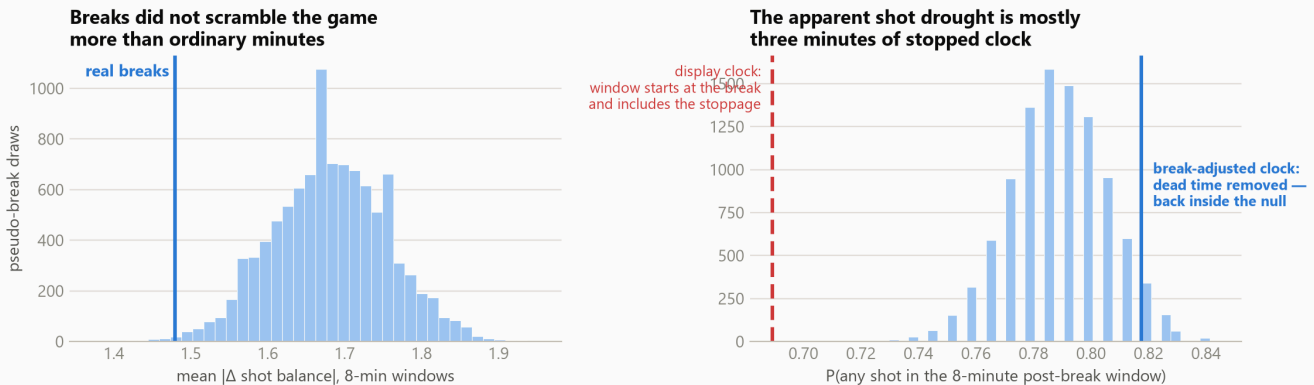
A note on what is not here. Per-shot xG does not exist in any independently auditable source for this tournament. FIFA's official post-match reports publish shot logs without xG; FBref blocks automated access. Outcomes are therefore shot-based (CHANGELOG A2). This is stated rather than papered over.

4. The seductive result, and its correction

Real hydration breaks vs 10,000 moments where nobody stopped the game

196 breaks; pseudo-breaks matched on half, score state and stage.

Post-break window measured two ways — display clock (starts at the break, includes the ~3-min stoppage) vs break-adjusted clock (dead time removed).



The outcome is the probability of any shot in the **8-minute post-break window** — and the two clocks define that window differently, which is the whole point:

- **Display clock:** the window begins when the break is called, so it *contains* the ~3-minute stoppage. This is what a viewer or an event feed sees.
- **Break-adjusted clock:** the hydration dead time is removed, so the window holds 8 minutes of displayed football.

On the display clock, the breaks look devastating. The probability of a shot in that window falls to **0.689**, against a null of **0.787** — far outside the range of ordinary matched minutes. If you stop there, you have a headline: hydration breaks kill momentum.

Remove the three dead minutes and it vanishes. On the break-adjusted clock the same probability is **0.818** against a null of **0.787** — squarely inside the null distribution. The apparent post-break shot drought was largely the stopped clock being counted as football. ("Momentum" in the broader sense — possession, territory, tactical control, psychology — is not what this measures; shots are.)

The primary metric agrees. The cleanest way to state it is the **paired effect**: for each break, its observed disruption minus the mean disruption across that break's *own* matched control minutes, averaged and bootstrapped by match.

Effect: -0.073 shots, 95% match-clustered CI [-0.258, +0.116] (183 breaks, 101 matches).

That interval includes zero. The point estimate is negative — real breaks were followed by marginally less disruption than comparable ordinary minutes — but the difference is not distinguishable from no difference once uncertainty is placed around the contrast itself and clustered by match. This is deliberately more conservative than the randomization percentile, which reflects only the variability of the controls, not the match-to-match variability of the real breaks.

Note what the interval still excludes: even its break-unfavourable end is **+0.07 shots** of extra disruption — orders of magnitude smaller than the decisive swings described publicly. The finding is *no detectable*

difference, not breaks calmed the game.

This estimate is not robust to demanding deeper control support. Requiring ≥ 3 , ≥ 5 and ≥ 10 clean control minutes per break moves it steadily more negative ($-0.073 \rightarrow -0.173 \rightarrow -0.196 \rightarrow -0.948$), and the ≥ 10 row excludes zero. That row rests on 18 breaks in 18 matches and is heavily selected — after the contamination fix, only quiet matches (few goals, cards or VAR stoppages, stable score state) can supply ten clean controls. The drift is consistent with selection on match character rather than a stronger break effect, but this design cannot separate them. The full-sample estimate is the headline; this caveat travels with it.

The average competitive effect is null. This replicates the existing preprint (arXiv 2607.19783) by an independent route.

What about the momentum graphic on screen?

Broadcasts showed a live momentum index all tournament, and it is what most viewers actually watched — so it deserves a direct answer rather than dismissal. Analysed with match-clustered uncertainty, treated breaks show a mean momentum drop of **7.91** against **7.11** at control minutes: a difference of **+0.80, 95% CI [-2.11, +3.85]**, straddling zero. Momentum does collapse after breaks — and collapses almost as hard after comparable ordinary minutes.

That comparison carries a **weaker design than anything else in this report**, for reasons that are properties of the data rather than choices. **No 2026 control minutes exist:** all 8,946 controls come from eleven other competitions (2018–2025), so the within-match pseudo-break design used for shots cannot be built for momentum at all. No minute-level momentum series is available either — only the pre/post values at each break survive, the curves having been stripped to avoid redistributing a proprietary index. And the index itself is an unpublished black box, which is why the specification keeps it strictly secondary.

Its value is triangulation. Two independent measurement systems, two different providers, two different control strategies — the same null. That agreement is worth more than either result alone. Full working and caveats: [reports/tables/provisional_null.md](#).

But that measures the match calming down — not momentum changing hands

Everything above measures **total** activity. Total activity can stay flat while momentum switches sides completely: if a team creating five shots to nil before the break creates one to four after it, the total is unchanged and the advantage has been wiped out. The public claim — *"we were all over them and the break killed it"* — is directional, so it needs a directional test. The frozen specification reserved one (Test B, spec §5): match on **pre-break dominance** and ask what normally happens after a comparable spell.

For each break we fix the attacking side from the pre-window alone (shots on target, ties on total shots), freeze that identity, and measure how much of its advantage survives. Controls are ordinary minutes in the same match and half, same score state, whose **own pre-window advantage equals the break's**. That matching is essential: teams that have just been dominating regress to the mean whether or not anyone stops the game, so an unmatched comparison would manufacture an effect from nothing.

window	breaks	attacking side's swing, real breaks	swing after comparable uninterrupted spells	difference	95% CI
5 min	88	-1.091	-0.955	-0.136	[-0.501, +0.247]
8 min	63	-1.032	-0.781	-0.251	[-0.775, +0.235]
10 min	33	-1.121	-1.379	+0.258	[-0.558, +1.034]

Among breaks with an identifiable and matchable attacking side, the attacking team lost roughly one shot of advantage afterwards — but no more than after comparable uninterrupted spells. We found no detectable additional loss associated with the hydration break.

That coverage qualifier matters: after requiring controls whose whole window clears the real break, only 88 / 63 / 33 of 203 breaks qualify at 5 / 8 / 10 minutes. The 10-minute row rests on ~30 matches and is underpowered rather than an independent confirmation. The rest are excluded because no side was clearly attacking in the pre-window, or because no exactly state-matched control minute exists. Both exclusions are principled and counted, but the conclusion covers the matchable subset, not every break.

Momentum changed hands (the attacking side ended up behind on shots) after **18–27%** of breaks, against **27–36%** of comparable ordinary spells — every interval spans zero. Shots-on-target swings show the same null. Tightening the match to require comparable *intensity* (equal shots on target, or equal total shots, or both) leaves every cell's interval spanning zero as well — but each tightening costs coverage, and the strictest cells fall to 14–27 breaks. Read those as coverage-limited, not as four independent confirmations.

This is the answer to the strongest version of the criticism, and it is more interesting than the flat-total result: the collapse fans remember genuinely occurs, and it is what dominance normally does anyway. Test B is descriptive by design — matching on pre-window dominance rebuilds part of the selection mechanism, which is why Test A is the primary matched counterfactual analysis. Full working, exclusions, weighting and the orientation diagnostic: [reports/tables/test_b.md](#).

A result we deleted

The analysis initially produced a strong directional finding: home teams gained shot differential after breaks (+0.29 after placement matching, with essentially no null draws reaching the observed value). It survived every robustness test applied to it.

It is not reported as a finding, because the control pool itself is biased for directional contrasts. All unfiltered eligible minutes average **+0.040** — essentially zero, as symmetry demands. The score-state-matched control pool averages **-0.134**, and is negative in *all three* score-state buckets. That is impossible for a clean control: a baseline tracking game state would be negative when the home side leads and sits back, and **positive** when it trails and chases. The cause is the preregistered goal-proximity screen, which strips disproportionately many home-pressure phases out of the controls while real breaks sample them at the natural rate.

The headline is unaffected — the primary metric is absolute, and the same bias runs *conservative* there. Full evidence: [reports/tables/robustness.md](#) ; limitation logged as CHANGELOG A5.

Does the null hold up?

Mostly, with one documented exception. All rows use the primary unmatched specification; placement matching is no longer computable against contamination-free controls and is reported as a coverage failure (CHANGELOG A7), not quietly dropped.

check	result
Paired effect, ≥ 3 / ≥ 5 / ≥ 10 control minutes	-0.173 / -0.196 / -0.948 — not stable; the ≥ 10 row excludes zero
Leave-one-match-out (102 refits)	gap stable in [-0.101, -0.044]; no single match drives it
Drop windows containing a red card	1.594 vs 1.661 — unchanged
Drop breaks with a goal in the prior 3'	1.494 vs 1.581 — unchanged
Nominal 22'/67' timing instead of actual	1.595 vs 1.606 — unchanged
Group vs knockout stage	1.622 vs 1.739 · 1.500 vs 1.474 — neither above null
Windows of 5 / 8 / 10 minutes	consistent throughout

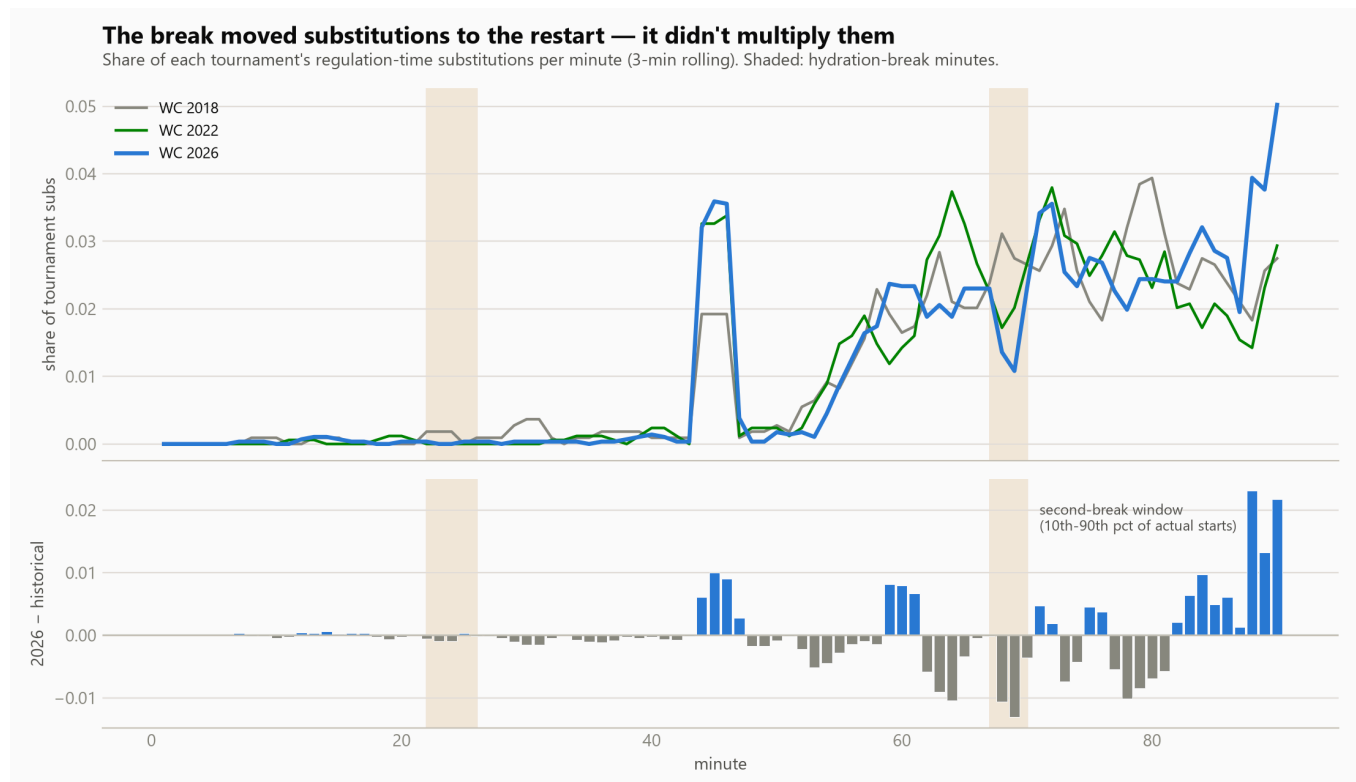
The first row is the honest weak point of this report, and §4 above states it in full: the estimate drifts as deeper control support is demanded, and the strictest cut — 18 breaks in 18 quiet, heavily selected matches — excludes zero. Everything else holds.

One exploratory exception, flagged and not claimed: the halves diverge. First-half breaks are markedly *less* disruptive than matched minutes (1.18 vs 1.50), while **second-half breaks are the one subgroup sitting above its null** (1.99 vs 1.81, pct 0.068 — not significant). That is coherent with the substitution evidence below — but it is one subgroup cut among several and needs preregistration before it can be a claim.

5. What did change

The competitive null is only half the story. Several things about World Cup football did measurably change.

Substitutions moved — they did not multiply



The intuitive test fails. Only **18.7%** of 2026 second-half substitutions fell within ± 3 minutes of their own match's break — *below* the minute-matched historical expectation (19.6% from 2018, 20.7% from 2022). The break did not become a substitution magnet.

The displacement pattern underneath it is the real finding (exploratory, not preregistered): a **deficit in the three minutes before the stoppage** (6.8% vs 7.6/8.5% historically) and a **surplus in the three minutes after the restart** (15.0% vs 12.6/14.6%). Coaches did not make more changes; they waited for the break and made them at the restart. The break became a scheduling point.

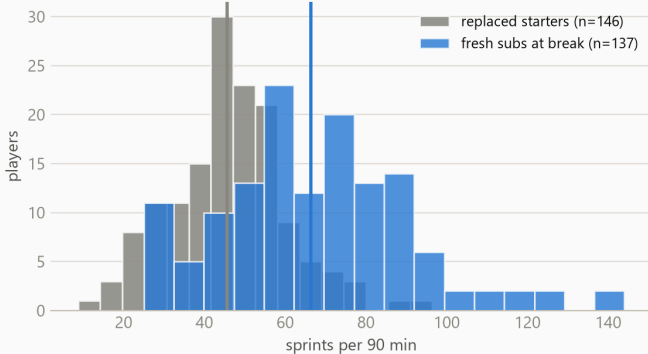
Context: 2026 averaged 4.64 regulation substitutions per team-match against 4.39 in 2022 and 2.87 in 2018 — the jump is the five-substitution rule, not the breaks. Median first substitution: 58' (2026), 57' (2022), 61' (2018).

The break as a fresh-power window

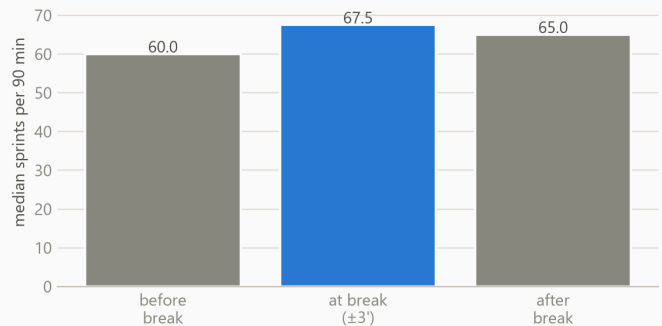
The second break as a fresh-power window

Per-90 physical output of WC2026 substitutes (FIFA match totals). Deployment, not physiology.

Fresh legs sprint far more than the legs they replace



Entry timing barely matters (matched appearance length)



FIFA's per-player physical data allows one legitimate question — not whether breaks refreshed players (that is unmeasurable here), but **whether coaches used them to inject fresh legs**.

Substitutes entering at the break ran at **66.3 sprints per 90** against **45.7** for the starters they replaced; high-speed running 162 vs 121, high-speed distance 1,245 m vs 806 m. Fresh legs are dramatically more explosive than tiring ones — as expected.

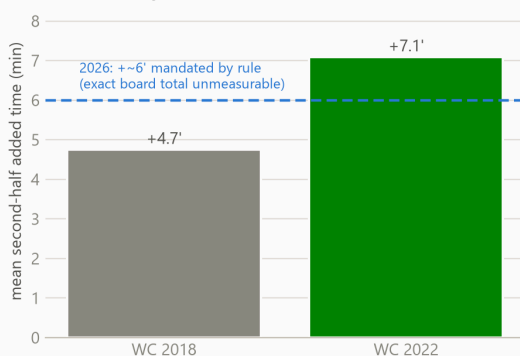
Honesty check: entry *timing* barely matters. Restricted to a matched ≤ 30 -minute appearance band, median sprints/90 are 60.0 before the break, 67.5 at it, 65.0 after — the timing differences largely collapse. The fresh-versus-tired gap is the story; the break's role is that it is where the swap gets coordinated.

Matches got longer — but 2022 got there first

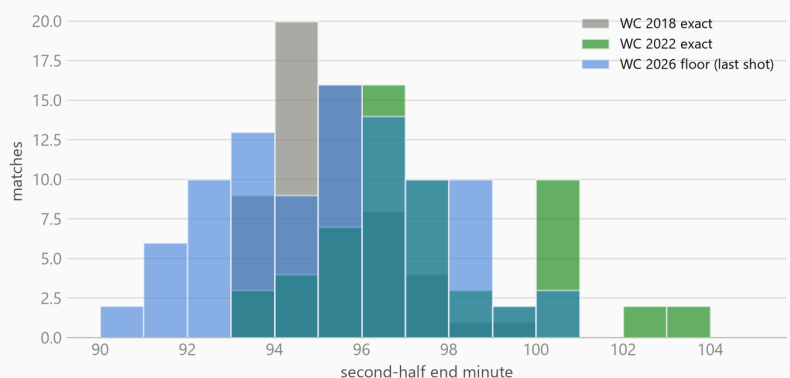
Did hydration breaks make matches even longer?

2018/2022 exact half-ends (StatsBomb); 2026 is a last-shot lower bound (exact board minutes unavailable).

2022 already stretched the clock



How late the second half ran



Qatar 2022 was the added-time anomaly: second halves ran to **97.1'** on average (+7.1), against **94.7'** (+4.7) in 2018. 2026 adds roughly six minutes per match **by rule**, since the breaks sit inside the clock.

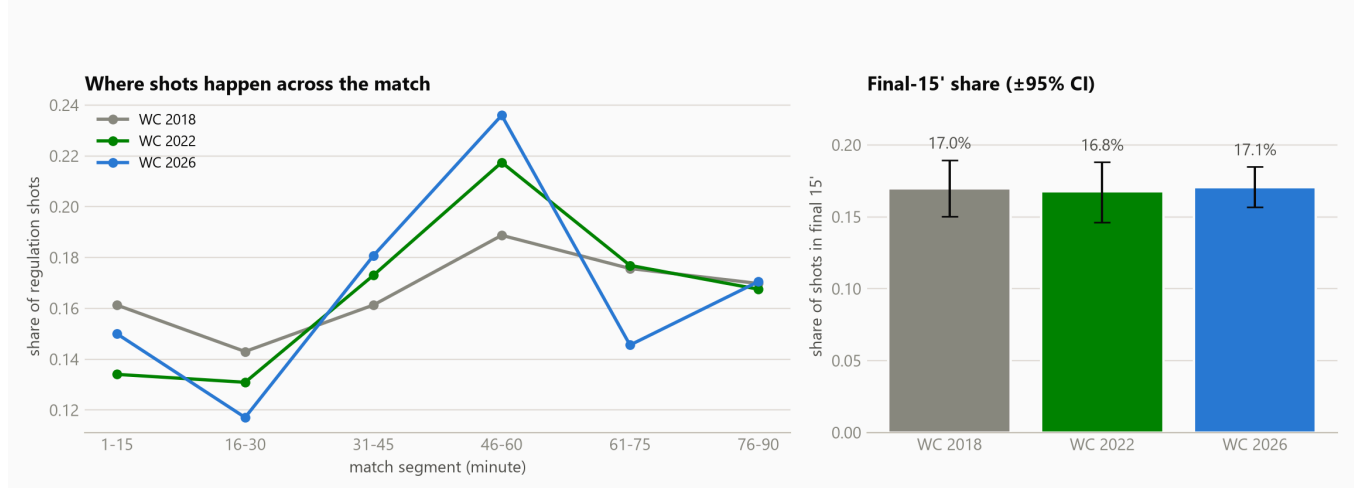
Whether 2026's boards *exceeded* the already-inflated 2022 environment cannot be determined: exact added-time minutes are absent from every auditable source we hold (the FIFA feed strips stoppage notation in 97 of

104 matches). We report a last-shot lower bound and no more. Late scoring shows no clean trend either — goals at $\geq 90'$ were 11.4% (2018), 8.9% (2022), 12.6% (2026).

Did the game stay alive later?

Did the game stay alive later once breaks existed?

Behavioral proxy — not physical. 2026-vs-2022 isolates the break era (both 5-sub); 2022-vs-2018 is the sub-rule jump.



The physical-recovery question — did breaks help players sustain the game? — cannot be answered directly: no per-player physical data exists for 2018 or 2022, and 2026's data are match totals with no pre/post-break split. (This was searched and confirmed, not assumed.)

The available proxy is behavioral: the share of shots in the final 15 minutes. It is **flat across all three tournaments** — 17.0% (2018), 16.8% (2022), 17.1% (2026), with overlapping confidence intervals. Whatever the breaks did, they did not visibly change how alive the closing stages were.

6. Why it felt different

If the average effect is null, the perception needs explaining. This is where the project departs from prior work.

We preregistered an inclusion rule — a claim counts only when a commentator, manager, player or major outlet attributes a change in a **specific match** to a **specific break** — then collected in two passes:

1. a **topical pass** across outlets writing about the breaks (Opta Analyst, Goal.com, Middle East Eye, The Times, TNT Sports, ESPN, England Football);
2. a **pre-specified stratified random sweep** of 24 matches / 48 breaks, each searched with an identical query template, logged in `data/manual/perception_sweep_log.csv` including every match where nothing was found.

The sweep is what makes the denominator credible. Searching only the matches *likely* to have claims would have rebuilt the exact bias under measurement, so the sample was fixed before any searching and every drawn match was searched whatever the result.

That yielded **22 usable claims and 24 logged rejections**. Every claim was then re-read against its source URL. Nineteen were confirmed at source; **three were not** and are excluded from all figures — one podcast that could never be located, one page that blocks automated access, and one quote absent from the page it was cited to. Two surviving claims needed correction: a break number (the source specified the second-half break, not the first) and a claim text (the source does not say the goal came a minute after the break; our own timeline puts them thirteen minutes apart). Both failures and both corrections originated in search-engine summaries that did not survive contact with the actual page — which is why the verification pass exists.

Each verified claim was then evaluated **blind to its text**: only the match, the break and the team said to benefit were read, and the swing was scored against that same match-half's own pseudo-break null. Several breaks drew claims from more than one outlet, so the statistically independent measure deduplicates to unique claimed breaks:

7 of 12 unique claimed breaks were supported (58%) — claim-level, for comparison, is 8 of 16 (50%). The two differ because heavily covered incidents contribute several rows; here that pulls the claim-level rate *down*, since the most-covered break (England–DR Congo's first, three separate claims) is not supported. Either way: the people making these claims were right about as often as not — and when right, often dramatically so.

The bias is not in the claims. It is in the **denominator**, and the random sweep measures it directly:

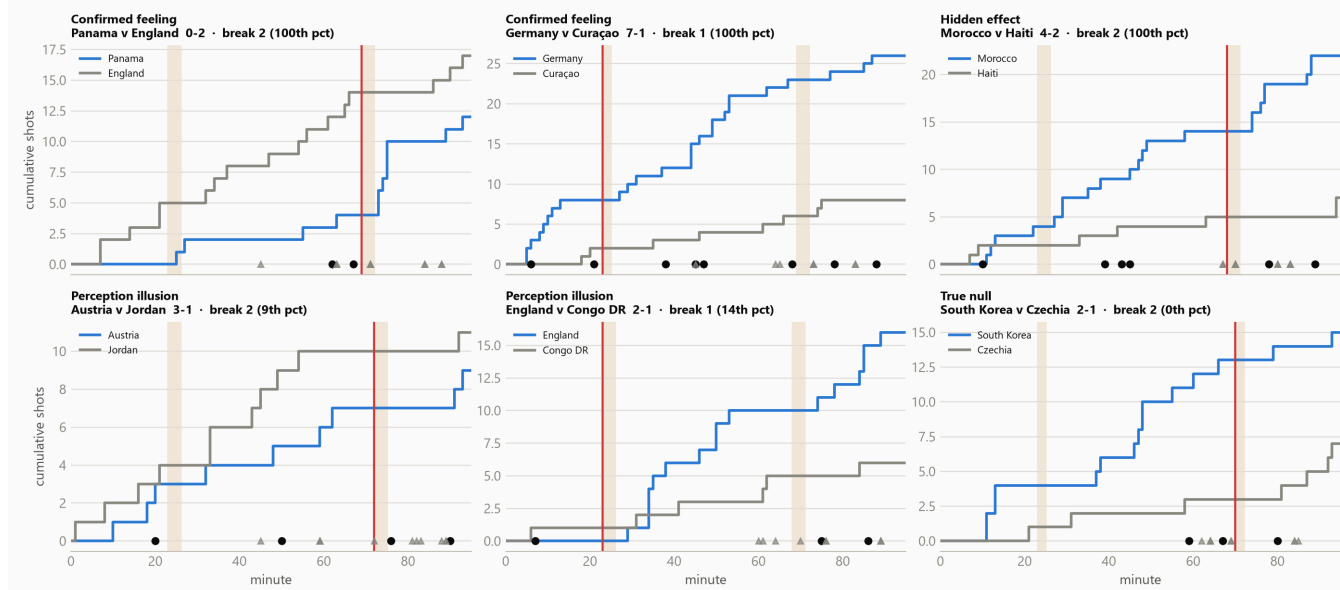
Public claims attached to just **4 of 48 randomly sampled breaks — 8.3%** (95% CI 3.3–19.6%), extrapolating to roughly **17 of 203 breaks** tournament-wide. Within that small, self-selected set the claims are right roughly half the time. Pundits were describing real swings. They were describing them from the tail of a distribution, and the tournament-wide story was then written from that tail.

The case-study matrix makes the asymmetry concrete. Scoring every break on swing size and crossing it with whether any claim was located:

	large swing	small swing
publicly claimed	3	4
no claim located	64	73

Six breaks, chosen by matrix — not by fame

Shaded = hydration breaks · red line = the break in question · dots = goals · triangles = substitutions on. Percentile is vs the same match-half's ordinary minutes.



Cases were selected by this matrix, not by fame. Germany–Curaçao qualified on merit at the 100th percentile. Netherlands–Sweden also scored at the 100th percentile but was displaced from the final six by Panama–England's larger swing; Switzerland–Bosnia, the third planning-stage favourite, never qualified at all. The hidden-effect slot went to Morocco–Haiti's second break — a 100th-percentile swing with no located commentary.

One case changed cells when the sweep ran, and it is worth reporting as such. Panama–England's second break is the tournament's largest pressure inversion: Panama went from 1 shot to England's 3 before it, to 6 against England's 0 after it — and still lost 0–2. The earlier topical pass found no commentary, so it was classified as a "hidden effect" and flagged *provisionally*, on the explicit ground that "no claim found" is not "no claim exists". The systematic sweep then located a TNT Sports data piece reporting precisely that dip. It is now a confirmed-feeling case. The caveat did its job, and the correction is left visible rather than quietly absorbed.

Meanwhile Austria–Jordan, which *did* generate public claims, sits at the **7th percentile** — the swing people remembered was smaller than a typical quiet passage of that same half.

Large post-break swings were common. Commentary about them was not. What determined whether a swing became a story was largely whether the scoreboard followed.

Coverage caveat: the sweep covers 24 of 104 matches. Its 8.3% rate is an unbiased estimate with a wide interval; `has_claim = False` for unswept matches means "no claim located", not "none exists".

7. Conclusion

Hydration breaks did not, on average, produce unusually large changes in shot balance. Once the stoppage itself was removed from the clock, post-break shooting activity looked ordinary. Measured against thousands of statistically comparable moments where nobody stopped the game, the football after a break

looks like ordinary football. The effect that seems obvious on a television clock is an artifact of counting three stopped minutes as play. This holds across windows, stages, exclusions, and every match left out in turn — with the one documented exception recorded in §4: requiring deeper matched-control support drives the estimate away from zero on a small, heavily selected subsample. The claim is *no detectable average effect on shot balance*, not *no effect under every specification*.

But the sport around the scoreboard did change. Two guaranteed tactical meetings per match became a fixed feature of coaching. Substitutions were re-timed to the restart — coordinated rather than multiplied. Matches carry roughly six mandated extra minutes on top of an already-stretched stoppage environment. And breaks became a scheduled window for injecting players who run half again as hard as the ones they replace.

And the perception was not a delusion — it was a sampling problem. When people made a specific, testable claim, the measured swing supported them roughly half the time, and sometimes very strongly. What never entered the conversation was the large majority of breaks that generated no public narrative at all — whether or not ordinary football volatility produced a swing afterwards. Our own case matrix contains dozens of breaks with swings as large as the famous ones and no located commentary. A rule applied 203 times was judged on the handful of occasions when the scoreboard happened to move immediately afterwards.

That is the honest shape of it:

The breaks did not consistently produce unusual swings in shot balance. They changed how coaches managed games, how long matches ran, how broadcasters packaged them — and they handed audiences a vivid, memorable, and deeply unrepresentative sample of evidence about themselves.

And the money

The loudest public charge was that FIFA interrupted football to sell advertising. Our data can measure the inventory, and cannot measure intent — so it does both, separately.

The inventory is real and quantifiable. The 203 recorded breaks account for **580 minutes — about 9.7 hours** of in-match stoppage, roughly 5.7 minutes per match; the policy guarantees **208 slots across all 104 matches**, which at the observed median duration would be **up to approximately 10.4 hours**. That upper figure is an arithmetic ceiling on stoppage time, not a measure of advertising sold: we did not audit broadcast schedules and do not know what share of it carried advertising, if any. Three properties make that commercially distinctive, and all three follow from the policy plus our own timing data: it is **guaranteed** (every match, both halves, irrespective of weather [1]), **predictable** (clustered at 23' and 68', so it can be sold in advance rather than filled reactively), and **inside the match** (the clock runs and the time is added back as stoppage time [1], so the audience does not disperse as it does at half-time).

Intent is not measurable here, and we do not claim it. No dataset in this project speaks to why the policy was adopted. FIFA presented it as player welfare [1], and its president publicly denied any revenue motive. Establishing motive would require internal deliberations or rights-negotiation records that we do not have; anyone asserting it — in either direction — is going beyond the public evidence.

What the data does add to the debate is this: the welfare rationale is weakest exactly where the commercial question is strongest. **Only 19% of breaks occurred at or above the threshold that made cooling breaks**

mandatory under FIFA's own previous protocol (§2). Four in five happened in conditions the old rules did not consider to require one. That does not demonstrate a commercial motive — FIFPRO's position is that the old threshold was simply too high [4] — but it is what makes the question legitimate rather than cynical.

FIFA called the breaks a player-welfare measure. The policy also created roughly ten hours of guaranteed, predictable, mid-match commercial inventory that did not previously exist. Both statements are true; neither establishes the other.

Full working: [reports/tables/commercial.md](#).

7b. Appendix — the one non-shot outcome, and why there is only one

A fair objection to everything above is that it rests on shots. If a break changed the *character* of a match — tempo, territory, structure — shot counts might not show it. We looked for other outcomes in the 2026 layer and found exactly one that is both time-stamped and dense enough to test.

Yellow cards, as a weak proxy. Differencing each break against its own matched control pool: $D = -0.0035 / -0.0054 / -0.0039$ at the 5 / 8 / 10-minute windows, **every interval including zero**. Booking rates after breaks are indistinguishable from those after matched ordinary minutes. Read this narrowly: a booking is a referee decision, driven by referee thresholds, tactical fouling, dissent, score state and match importance. Card rates can move without the run of play changing, and vice versa. At roughly 0.16 cards per 8-minute window only a very large shift would be detectable. This is weak evidence about **booking rates**, not a finding that the breaks left the game's character unchanged.

Goals cannot be used at all, and the reason matters. Goals are denser than cards (308 vs 253) and look like the better outcome. But Test A's preregistered screen excludes candidate minutes within 3 minutes of a goal, so control windows are goal-depleted *by construction*: 0.227 goals per 8-minute break window against 0.054 in matched controls. That four-fold gap is the screen, not the breaks. Reporting it would have produced a dramatic false positive — one we caught only because the mechanism was checked before the number was believed.

There is no positional or build-up version of this table, and none is possible here. FIFA's post-match reports do contain rich tactical data — line height, team length, build-up phases, line breaks, passing networks, defensive pressure. Every page of six such reports (~300 pages) was checked for a half split, a minute bin or any per-period breakdown: **none exists**. Those metrics are one value per team per match. A before/after-break comparison of them is not a hard analysis; it is an undefined one. StatsBomb's 2018 and 2022 data does carry true event coordinates — but those tournaments had no universal breaks, so there is nothing to compare against.

Full working: [reports/tables/cards.md](#).

8. Limitations

Stated plainly, because they bound every claim above.

1. **No per-shot xG** in any auditable source; outcomes are shot-based.
2. **Home/away-oriented results are not reportable** — the control pool is biased for that particular signed contrast (CHANGELOG A5). Test B is directional but avoids it by orienting on pre-window dominance, symmetrically in both arms; its home/away split is shown as a diagnostic only and is not a finding. Test B is also descriptive by design, not causal.
3. **2026 added time is unmeasurable** at board level; a lower bound is reported instead.
4. **Physiology is out of scope.** No claim here concerns player hydration, core temperature, or injury. Public event data cannot support them.
5. **Perception collection covers 24 of 104 matches** by random sweep (plus a topical pass): 22 claims, of which 19 were confirmed at source and 3 could not be. Only confirmed claims are used in any figure. One source (ESPN) blocks automated access and its claim awaits a manual browser check; two others should be treated as withdrawn unless a primary source is produced.
6. **Coverage is 102 of 104 matches, 203 of a possible 204 breaks.** Every exclusion is listed with a reason in `data/processed/exclusions.csv`.
7. **Substitution minutes in the 85–90' tail are not comparable across sources** — a recording artifact, flagged and not interpreted.
8. **Everything here is association**, not causation. Breaks were not randomly assigned; the placebo design asks whether post-break football was unusual, not whether the break caused it.

9. Reproducing this

```
pip install -r requirements.txt
python -m src.audit           # source inventory + hashes
python -m src.build_tables    # master tables
python -m src.placebo         # confirmatory placebo (Core Output 3)
python -m src.robustness      # robustness pass
python -m src.subs            # substitution timing (Core Output 4)
python -m src.physical        # fresh legs
python -m src.late_game       # late-game proxy
python -m src.added_time      # added time (Core Output 6)
python -m src.perception      # perception evaluation (Core Output 5)
python -m src.case_studies    # case selection (Core Output 7)
python -m src.test_b          # Test B: state-matched directional (spec $5)
python -m src.break_window    # clock-artefact analysis (exploratory, E1)
python -m src.cards           # appendix: yellow-card rates
python -m src.commercial      # commercial inventory arithmetic
python -m src.facts           # regenerate reports/facts.json + FACTS.md
python -m pytest tests -q      # incl. prose-vs-numbers sync checks
```

Every headline number in this report is generated into `reports/FACTS.md` by `python -m src.facts`, and `tests/test_report_sync.py` fails the build if this document drifts from it.

Sources, commits and SHA-256 hashes: `config/sources.yaml` , `data/processed/source_inventory.csv` .
Frozen specification: `spec.md` . All amendments: `CHANGELOG.md` .

10. References

Policy and rules

[1] FIFA (primary source). *Players to benefit from hydration breaks at FIFA World Cup 2026*. States that the referee stops play 22 minutes into each half, that each break lasts up to three minutes, that the time is **added back as stoppage time**, and that the breaks apply in all matches with no weather or temperature condition attached. <https://inside.fifa.com/organisation/news/hydration-breaks-world-cup-2026-player-welfare> Contemporary reporting and criticism: https://www.espn.com/soccer/story/_/id/48945011/why-there-drinks-breaks-2026-world-cup-fifa-criticised

[2] IFAB, *Laws of the Game*, Law 7 (The Duration of the Match). Distinguishes a **drinks break** (≤ 1 minute) from a **cooling break** (90 seconds to 3 minutes), and requires allowance for time lost. <https://www.theifab.com/laws/latest/the-duration-of-the-match/>

[3] FIFA heat protocol (pre-2026): cooling breaks mandatory at WBGT ≥ 32 °C, requestable roughly 27–32 °C. This project did not obtain FIFA's protocol document directly; the threshold is taken from secondary reporting and should be treated as such — <https://kestrelinstruments.com/blog/fifas-hydration-break-rule-explained-what-it-means-for-the-2026-world-cup> and the FIFA–FIFPRO threshold dispute reported at https://www.espn.com/soccer/story/_/id/49343877/fifa-world-cup-fifpro-temperature-heat-protocol-talks-sources. Anyone citing the 32 °C figure in a publication should verify it against FIFA's own medical/competition regulations.

[4] FIFPRO heat-stress guidance: breaks recommended from ~ 26 °C WBGT; match delay advised above 28 °C. <https://fifpro.org/en/supporting-players/health-and-performance/extreme-weather-and-climate-change/> FIFA–FIFPRO threshold disagreement reported at https://www.espn.com/soccer/story/_/id/49343877/fifa-world-cup-fifpro-temperature-heat-protocol-talks-sources

Prior work

[5] *Do In-Match Hydration Breaks Alter Match Momentum? A Within-Match Case-Crossover Analysis of the 2026 FIFA World Cup*. arXiv:2607.19783. <https://arxiv.org/html/2607.19783>

[6] Opta Analyst, *Are World Cup Hydration Breaks Really Killing Momentum?* <https://theanalyst.com/articles/world-cup-2026-hydration-breaks-momentum-stats>

Data sources

[7] FIFA Post Match Summary Reports (PMSR), fifatrainingcentre.com — shot logs, substitutions, per-player physical data, team statistics. Retrieved 2026-07-24; per-file hashes in `data/processed/source_inventory.csv` .

[8] StatsBomb Open Data — World Cup 2018 and 2022 events (substitutions, shots, goals, period ends). <https://github.com/statsbomb/open-data>

[9] Ddey07/wc2026-hydration-momentum — break timings, treated/control covariates, SofaScore-derived momentum summaries. <https://github.com/Ddey07/wc2026-hydration-momentum>

[10] mominullptr/FIFA-World-Cup-2026-Dataset — tournament metadata, venues with coordinates, lineups. <https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset>

[11] anup4khandelwal/wc26-analytics — PMSR parser approach (code reference only; no numbers used). <https://github.com/anup4khandelwal/wc26-analytics>

Perception sources

Full per-claim source URLs, verification status and rejection reasons: `data/manual/perception_claims.csv`, `perception_rejections.csv`, `perception_sweep_log.csv`. Outlets cited: Opta Analyst, Goal.com, Middle East Eye, The Times (secondhand), TNT Sports, England Football.

Commercial context

[12] Reporting on hydration-break advertising inventory and World Cup broadcast revenue, e.g. <https://fortune.com/2026/03/09/world-cup-hydration-breaks-tv-ads-revenue-fifa-global-viewership/> and https://www.espn.com/soccer/story/_/id/49326566/hydration-breaks-world-cup-soccer-broadcast-rights

Note on provider data: raw provider files (FIFA PMSR PDFs, SofaScore-derived arrays) are **not redistributed** in this repository. Only project-generated analytical tables are committed. See `LICENSE` and the data-availability note in `README.md`.